

Zimu Yu

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EDUCATION

Zhongnan University of Economics and Law (ZUEL)

Sep 2021 – Jul 2025

Major: Bachelor of Science in Statistics **Average:** 87/100

Core coursework: Probability Theory, Mathematical Statistics, Multivariate Statistical Analysis, Statistical Learning.

RESEARCH INTERESTS

Machine learning; computational biology; protein variant effect prediction; mutational epistasis; protein engineering.

PUBLICATIONS AND MANUSCRIPTS

* Equal contribution.

- [1] W. Wang*, **Z. Yu***, et al. From single-sequence structure prediction to protein fitness landscapes through a composable, epistasis-aware mutation atlas. *bioRxiv* preprint, 2026; under review at *Nature Machine Intelligence*. [\[Preprint\]](#) [\[Code\]](#)
- [2] K. Xu*, **Z. Yu***, et al. RADAR enables marker-free fine-grained discovery of anomalous cells in multi-sample and multimodal single-cell omics. Under review at *Bioinformatics*. [\[Code\]](#)
- [3] W. Wang*, S. Yu*, Y. Shen, **Z. Yu**, et al. Deciphering the cellular sequence–metabolite synthetic landscape enables predictive design by MetaAI. Under review at *Nature Microbiology*. [\[Code\]](#)
- [4] K. Xu, Y. Du, M. Liu, **Z. Yu**, et al. Causality-Induced Positional Encoding for Transformer-Based Representation Learning of Non-Sequential Features. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*, 2025. [\[Paper\]](#) [\[Code\]](#)

PATENTS

- [1] **Z. Yu**, Y. Xu. Methods and systems for multi-site mutation effect prediction in biological sequences. Patent application submitted, 2026.

RESEARCH EXPERIENCE

Tsinghua University

Beijing, China

Research Assistant | Advisor: [Prof. Haipeng Gong](#)

Aug 2025 – Present

School of Life Sciences

Cerebra-Epistasis

(Publication [1])

- Proposed a composable, structure- and epistasis-aware mutation atlas approach for arbitrary-order multi-mutant fitness prediction and interpretable fitness landscape modeling.
- Designed and implemented an end-to-end framework jointly optimizing a single-sequence structure predictor and an SE(3)-equivariant fitness model to learn single-mutation effects and epistatic representations.
- Designed and conducted comprehensive evaluations across diverse protein fitness benchmarks, demonstrating state-of-the-art performance in multi-mutant prediction, higher-order extrapolation, unseen epistasis prediction, and protein stability ($\Delta\Delta G$) prediction.
- Enabled efficient fitness landscape prediction by reusing the mutation atlas, achieving estimated speedups exceeding three orders of magnitude.

Zhongnan University of Economics and Law
Research Assistant | Advisor: [Prof. Yunxin Xu](#)
School of Statistics and Mathematics

Wuhan, China
Mar 2025 – Present

MetaAI

(Publication [3])

- Proposed and implemented an arbitrary-order epistasis modeling method for multi-mutant fitness prediction.
- Designed a higher-order factorization framework to capture epistasis and guide CHS and 4CL engineering.
- Co-designed and conducted a Low- N benchmark across public protein datasets to evaluate extrapolation from sparse low-order measurements to unseen higher-order mutants.

Protein Stability Prediction

(Manuscript in preparation)

- Curated large-scale experimentally measured $\Delta\Delta G$ datasets and developed a protein stability prediction model for estimating mutation effects.
- Introduced a confidence-filtered self-distillation strategy by adding high-confidence predictions back into model training, and selected high-confidence candidates for wet-lab validation.

Emory University

Research Assistant | Advisor: [Prof. Xiaobo Sun](#)
Department of Human Genetics

Atlanta, GA
Jan 2026 – Present

RADAR

(Publication [2])

- Proposed RADAR, a unified marker-free framework for anomalous-cell detection, cross-sample alignment, and fine-grained subtype resolution.
- Designed reconstruction-GAN and transferring-GAN modules for anomalous-cell detection and cross-sample alignment while preserving biological heterogeneity.
- Led comprehensive benchmarking across scRNA-seq, scATAC-seq, and spatial transcriptomics datasets spanning diverse tissues, diseases, species, and sequencing platforms.

TECHNICAL SKILLS

Programming Languages: Python, R; **Deep Learning Frameworks:** PyTorch; **Computational Biology:** PyMOL, Scanpy, Biopython; **Tools:** Linux, Git, LaTeX.